

COURSE SYLLABUS

STAT451 – Applied Statistics for Engineering and Scientists

1. GENERAL INFORMATION

Course name:	Applied Statistics for Engineering and Scientists
Course name (in Vietnamese):	Thống kê ứng dụng trong kỹ thuật và khoa học
Course ID:	
Knowledge block:	
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	MTH251 – Calculus 1; MTH252 – Calculus 2
Prior course:	
Instructors:	Dr. Hoàng Văn Hà

2. COURSE DESCRIPTION

The course is designed to provide students statistical analysis emphasizing engineering applications. Topics include descriptive statistics, probability, sampling distributions, estimation, hypothesis testing, regression, quality control, and reliability.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Compute probabilities of occurrences in well-defined probability spaces.	
G2	Model the event of random phenomena using discrete/continuous random variables.	

G3	Obtain sample statistics, such as sample mean and sample variance, from a data set.	
G4	Approximate the distribution of the sample mean using the central limit theorem.	
G5	Estimate unknown parameters using point/interval estimators.	
G6	Test the plausibility of a statistical hypothesis based on information from a data set.	
G7	Perform linear regression analysis on a data set.	

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1		
G1.2		

5. TEACHING PLAN

THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Chapter 1: Overview and Descriptive Statistics 1.1 – Basic statistical concepts 1.2 – Methods in descriptive statistics 1.3 – Measures of location 1.4 – Measures of variability		Lecturing Q&A, Group discussion, Demonstration	Quiz 1

2	Chapter 2: Probability 2.1 – Sample spaces and Events 2.2 – Properties of Probability 2.3 – Counting Techniques 2.4 – Conditional Probability 2.5 – Independence		Lecturing Q&A, Group discussion, Demonstration	Quiz 2
3	Chapter 3: Discrete Random Variables and Probability Distributions 3.1 – Random Variables 3.2 – Probability distribution for discrete RVs 3.3 – Expected Values 3.4 – Binomial distribution 3.5 – Poisson distribution		Lecturing Q&A, Group discussion, Demonstration	HW1
4	Chapter 4: Continuous Random Variables and Probability Distributions 4.1 – Probability density function 4.2 – Cumulative distribution function and expected values. 4.3 – Normal distribution 4.4 – Exponential distributions 4.5 – Other continuous distributions		Lecturing Q&A, Group discussion, Demonstration	HW2
	Midterm Exam			
5	Chapter 5: Sampling Distributions 5.1 – Sampling distributions 5.2 – Distribution of sample mean 5.3 – Distribution of sample variance 5.4 – Distribution of sample proportion		Lecturing Q&A, Group discussion, Demonstration	

6	Chapter 6: Confidence Intervals Based on a Single Sample 6.1 – Confidence interval for mean 6.2 – Confidence interval for variance and standard deviation 6.3 - Confidence interval for proportion		Lecturing Q&A, Group discussion, Demonstration	Quiz 3
7	Chapter 7: Hypothesis testing Based on a Single Sample 7.1 – Hypotheses & test procedures 7.2 – Test about a population mean 7.3 – Tests about a population proportion 7.4 – p-value method		Lecturing Q&A, Group discussion, Demonstration	Quiz 4
8	Chapter 8: Hypothesis testing Based on a Two Samples 8-1 – Hypothesis tests on the difference in mean of two normal distributions, variances known 8-2 – Hypothesis tests on the difference in mean of two normal distributions, variances unknown 8-3 – Hypothesis tests on the difference in population proportions		Lecturing Q&A, Group discussion, Demonstration	HW3
9	Chapter 9: Simple Linear Regression and Correlation 9-1 – Simple linear regression 9-2 – Properties of the Least Squares Estimators 9-3 – Correlation		Lecturing Q&A, Group discussion, Demonstration	HW4

LABORATORY

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the

corresponding topic and ask students to do their lab exercises either on computer in the lab or at home. All the lab work submitted will be graded. There would be a final exam for lab work.

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			30%
A11	Quizzes: Quiz 1 to Quiz 4	Small quizzes in class for each topic		8%
A12	Homeworks: HW1 to HW4	Practicing based on knowledge taught in class		12%
A13	Weekly lab work/exercise: LW1-LW10			10%
A2	Exams			70%
A21	Midterm exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems		30%
A22	Final exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems		40%

7. RESOURCES

Textbooks

- *Probability and Statistics for Engineering and the Sciences* (7th edition) by Jay Devore, Brooks/Cole, Cengage Learning, 2009.
- *Probability and Statistics for Computer Scientists* (2nd edition) by Michael Baron, CRC Press, Taylor and Francis Group, 2014.

Others

- Montgomery, Douglas C., and George C. Runger. *Applied statistics and probability for engineers*. John Wiley & sons, 2020.

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- Mobile phones, etc. must be silenced during all classroom lectures. Those not heeding this rule will be asked to leave the classroom immediately so as to not disrupt the learning environment.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.